Cover Letter — Electrical Engineering (Archiv für Elektrotechnik, Springer Nature)

Date: November 1, 2026
To: Prof. Dr. Igor Kuzle, Editor-in-Chief, Electrical Engineering (Archiv für Elektrotechnik)
Faculty of Electrical Engineering and Computing, University of Zagreb, Croatia

Subject: Submission of Original Research Article (via Springer Nature Transfer Desk):
"PSO-TCWavKAN: A Swarm-Optimized Temporal Convolutional Wavelet Kolmogorov-Arnold Network with Multi-Kernel Trend Decomposition and Asymptotic Linear Lower-Bound Guarantees for Resilient Power Transformer Thermal Dynamics Forecasting"

Dear Prof. Dr. Igor Kuzle and Esteemed Editorial Board,

I am pleased to submit my original research manuscript entitled "PSO-TCWavKAN: A Swarm-Optimized Temporal Convolutional Wavelet Kolmogorov-Arnold Network with Multi-Kernel Trend Decomposition and Asymptotic Linear Lower-Bound Guarantees for Resilient Power Transformer Thermal Dynamics Forecasting" for consideration as an Original Research Paper in Electrical Engineering (Archiv für Elektrotechnik).

Scientific Context & Alignment with Electrical Engineering:

Power transformer thermal forecasting is indispensable for Dynamic Thermal Rating (DTR), condition-based maintenance, and automated cooling control. In accordance with IEEE Std C57.91, IEC 60076-7, and IEEE Std C57.165-2024, each sustained 6 °C elevation in winding hot-spot temperature (HST) doubles insulation loss-of-life. Our manuscript explicitly distinguishes Top-Oil Temperature (OT) from HST: while fiber-optic HST sensors are rare, substation SCADA telemetry records bulk dielectric OT, which provides the foundational thermodynamic boundary condition for real-time differential HST calculations. This work advances the landmark wavelet forecasting methodology published in this journal by Bezerra et al. (2024) (Electrical Engineering, 106(3):3017–3029, DOI: 10.1007/s00202-023-02134-5) by resolving the discrete wavelet boundary limitations and standard MLP bottlenecks.

Five Transformative Scientific Contributions:

1. Resolution of KAN Grid Update Dilemma: Replaces piecewise B-spline knot grids with a continuously differentiable (C^infinity) dyadic Morlet wavelet filter bank across K_w = 5 octaves, eliminating discrete knot reallocation and avoiding optimizer momentum shocks by construction.

2. Multiscale Time-Frequency Representation: Achieves the Heisenberg-Gabor lower bound (Delta t * Delta omega = 1/2), capturing slow bulk oil thermal capacitance (tau_oil approx 1.5–3.0 h) and rapid transient load surges (tau_wind approx 4–10 min).

3. Decisive Superiority in Volatile Regimes & Theorem 1: Secures the lowest MSE in 7 of 8 lookback-horizon configurations on volatile ETTh2 (reducing MSE vs DLinear by up to 13.6% at H=720). Theorem 1 formally proves asymptotic linear subspace containment: the topological closure contains canonical DLinear (K1=K2=25) in the zero-gate limit, retaining linear fidelity while dynamically routing non-linear residuals.

4. Edge SCADA Efficiency & Low Latency: Requires only 44,841 parameters (3.8x reduction over PatchTST Lite) and achieves an inference latency of 3.4 ms with 3.3x faster training than standard B-spline KAN, enabling real-time edge execution on substation RTUs.

5. Bounded CLPSO Automated Co-Design: A constriction-factor swarm optimizer (chi = 0.7298, c1=c2=2.05) searches a 4-D parameter space in exactly 25 surrogate evaluations, bypassing non-convex ripple traps prior to AdamW fine-tuning and SWA.

Empirical Rigor & Statistical Framing:

Across all 16 canonical benchmark configurations (5 seeds for DLinear, NLinear, PSO-TCWavKAN; 3 seeds for PatchTST Lite), two-sided Wilcoxon signed-rank tests demonstrate a nominal uncorrected improvement over NLinear (W=28.0, p=0.0386), which attenuates to non-significance under Holm-Bonferroni correction (p_{Holm} = 0.116), with parity vs DLinear (p = 0.274) and PatchTST Lite (p = 0.0934). The manuscript candidly presents regime-specific dominance in non-stationary conditions rather than overclaiming unconditional statistical superiority.

Publication Model Declaration:

In accordance with Springer Nature policies for hybrid journals, this manuscript is submitted with the post-acceptance preference for the traditional subscription publishing route (without Open Access fees / 0 APC).

Data Availability & Ethical Harmonization:

All benchmark data are open-source. The complete v9.1 implementation, aggregate CSV matrices, statistical reproduction scripts, and plotting code are archived publicly on GitHub (https://github.com/nphu211206/KAN-PSO-Transformer-Forecasting) under the MIT license. All ethical declarations harmonize 100% verbatim with the manuscript.

Thank you for your time, consideration, and leadership of Electrical Engineering.

Sincerely yours,
Nguyen Phu Toan

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